Compare fuels such as methanol, ethanol, propanol and butanol to find out which is the 'best' fuel. Predict which one you think is the best.

Fill in the table below

Name of fuel			
Mass of water (g)			
Start temperature of water (°C)			
End temperature of water (°C)			
Temperature rise (°C)			
Start mass (fuel + burner) (g)			
End mass (fuel + burner) (g)			
Mass of fuel used (g)			
Energy transferred (J) *= mass of water (g) x 4.2 x temperature rise (°C)*			
Energy transferred per gram of fuel used (J/g) *= energy transferred (J) ÷ mass of fuel used (g)*			

Compare the results

1. Which is the best fuel?
2. Does this match your prediction?

Teaching notes

The sheet can be used to record results from an investigation to compare fuels.

The table could be part filled with some example results so that students can complete the table and calculate the energy transferred per gram of fuel.